

Unit 2 : Physical Layer

❖ Data and Signals

In data communication, data is the information to be transmitted (text, numbers, images, audio, video), and a signal is the physical representation of data that travels through a transmission medium. Data cannot be transmitted directly; it must first be converted into signals. Signals can be of two main types: analog signals and digital signals.

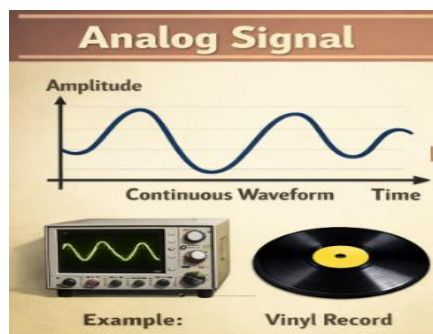
○ Analog and Digital Signals

Data can exist in two forms: Analog and Digital.

Analog data is transmitted in a continuous form, whereas digital data is transmitted in a discrete form.

• Analog Signals

Analog signals are continuous waves that vary smoothly over time. They are transmitted in the form of waveforms.



Key Points:

- Analog signals are continuous in nature.
- They are visualized as a smooth curve (oscillation).
- They are transmitted through a medium by generating smooth, rolling cycles.
- An analog signal is defined using amplitude, frequency, and phase.
- They can be captured using devices like a microphone.
- Analog signals are not resistant to noise, so they suffer from distortion and reduced transmission quality over long distances.

Examples:

- Human voice
- Sound waves
- Temperature signals
- Traditional radio signals

• Digital Signals

Digital signals are discrete signals that transmit data in the form of binary values (0 and 1). They are usually represented by square waves.



Key Points:

- Digital signals transmit data as ON/OFF or 1/0.
- They are defined by bit interval and voltage levels.
- Digital data can be stored in memory easily.
- They are more resistant to noise, so distortion is very less.
- For long-distance transmission, digital signals are regenerated using repeaters.
- They transmit data in limited, fixed values.
- Digital signals require higher bandwidth compared to analog signals.

Examples:

- Computer data
- Mobile communication
- Internet signals
- Digital audio and video

❖ Key Terms: Period, Frequency, Phase, Bandwidth, Bit rate

1. Period (T)

- The period is the time required to complete one full cycle of a signal.
- It is measured in seconds (s).
- Represents how long one waveform repetition takes.

Formula:

- $T = 1/f$ or $f = 1/T$
- Where f = frequency

Example:

If frequency = 50 Hz, then
 $T = 1/50 = 0.02$ seconds.

2. Frequency (f)

- Frequency is the number of signal cycles transmitted in one second.
- Measured in Hertz (Hz).
- Indicates how fast a signal repeats.

Formula:

- $f = 1/T$ or $T = 1/f$
- Higher frequency means more data can be transmitted.

3. Phase

- Phase describes the position of a waveform relative to time zero or another signal.
- Measured in degrees (°) or radians.
- Indicates how much a signal is shifted forward or backward.

Example:

Two signals starting at different times have a phase difference.

4. Bandwidth

- Bandwidth is the range of frequencies a communication channel can transmit.
- Measured in Hertz (Hz).
- Determines the data carrying capacity of a channel.

Formula:

- $\text{Bandwidth} = f_{\text{high}} - f_{\text{low}}$
- Greater bandwidth $\Rightarrow$ higher data transmission capacity.

5. Bit Rate

- Bit rate is the number of bits transmitted per second.
- Measured in bits per second (bps).
- Shows speed of data transmission.
- Example:
1 Mbps = 1,000,000 bits per second.
- Higher bit rate means faster communication.

❖ **Transmission Impairment**

Transmission impairment refers to the degradation of signal quality while it travels through a communication medium.

During transmission, signals may lose strength or become altered due to various factors.

The main types of transmission impairment are:

- Attenuation
- Distortion
- Noise

➤ **Attenuation**

Attenuation is the loss of signal strength as a signal travels through a transmission medium from sender to receiver. As distance increases, the signal becomes weaker due to resistance and energy loss in the medium.

Attenuation refers to the reduction in amplitude or power of a signal during transmission over a communication channel.

Causes of Attenuation

- Resistance of cables and transmission medium
- Signal energy converted into heat
- Long transmission distance
- Imperfections in the communication medium

Measurement

- Attenuation is measured in decibels (dB).
- It indicates how much signal power has been reduced.

Effects of Attenuation

- Weak signal at receiver
- Data transmission errors
- Poor voice or internet quality
- Reduced communication reliability

Solution / Prevention

- Use amplifiers to increase signal strength.
- Use repeaters to regenerate signals over long distances.
- Use high-quality transmission media.

Example

A telephone or internet signal becomes weak when transmitted through very long cables due to attenuation.

➤ **Distortion**

Distortion is a type of transmission impairment that occurs when the shape or form of a signal changes during transmission through a communication medium. As a result, the received signal becomes different from the original transmitted signal.

Distortion refers to the alteration of a signal waveform caused by different frequency components traveling at different speeds in the transmission medium.

Causes of Distortion

- Different frequencies travel at unequal speeds.
- Unequal delay characteristics of the communication channel.

- Imperfections in transmission media.
- Bandwidth limitations.

Working Principle

A signal is made up of multiple frequency components. When these components arrive at the receiver at different times, the signal shape changes, causing distortion.

Effects of Distortion

- Received signal differs from transmitted signal.
- Errors in data interpretation.
- Poor audio or video quality.
- Loss of communication accuracy.

Prevention / Solution

- Use transmission media with uniform delay characteristics.
- Increase channel bandwidth.
- Use equalizers to correct signal distortion.

Example

During a phone call, the voice may sound unclear or altered because the signal experiences distortion during transmission.

➤ Noise

Noise is any unwanted electrical or electromagnetic signal that interferes with the original transmitted signal during communication. Noise disturbs the signal and reduces the accuracy and quality of data transmission.

Noise refers to undesired disturbances that affect a signal while it travels through a transmission medium, causing errors at the receiver.

Causes of Noise

- Electrical devices and machines
- Natural atmospheric disturbances
- Heat generated in electronic components
- External electromagnetic interference

Types of Noise

1. Thermal Noise
 - Caused by random motion of electrons due to heat.
 - Present in all electronic devices.
 - Cannot be completely eliminated.
2. Induced Noise
 - Occurs due to interference from nearby electrical equipment such as motors and power lines.
 - Also called electromagnetic interference.
3. Crosstalk
 - Happens when signals from one communication channel interfere with another adjacent channel.
 - Common in telephone cables.
4. Impulse Noise
 - Sudden and irregular disturbances.
 - Caused by lightning, switching operations, or power surges.
 - Produces sharp spikes in the signal.

Effects of Noise

- Data transmission errors
- Loss of signal clarity
- Reduced communication quality
- Need for retransmission of data

Prevention / Control Methods

- Proper shielding of cables
- Use of filters
- Error detection and correction techniques
- High-quality transmission media

❖ Performance

Performance in data communication refers to how efficiently and quickly data is transmitted from sender to receiver over a network. Network performance is evaluated using parameters such as Throughput, Latency, Transmission Time, and Queuing Time.

➤ Throughput

Throughput is the actual amount of data successfully transmitted through a communication network in a given period of time. It represents the real data transfer rate achieved during communication.

Throughput refers to the number of bits received successfully at the destination per second.

Measurement

- Measured in bits per second (bps).
- Common units include:
- Kbps (Kilobits per second)
- Mbps (Megabits per second)
- Gbps (Gigabits per second)

Characteristics of Throughput

- Indicates real network performance.
- Usually less than bandwidth.
- Affected by network traffic and delays.
- Depends on hardware, protocols, and transmission medium.

Factors Affecting Throughput

- Network congestion
- Transmission errors
- Latency and delay
- Packet loss and retransmission
- Quality of network devices

Effects of Low Throughput

- Slow file downloads
- Video buffering
- Poor communication quality

Example

If a network has a bandwidth of 10 Mbps, but due to congestion only 7 Mbps data is successfully delivered, then the throughput is 7 Mbps.

➤ Latency

Latency is the total time taken by data to travel from the sender to the receiver across a communication network. It represents the delay experienced during data transmission.

Latency refers to the time delay between sending data and receiving it at the destination.

Measurement

- Measured in milliseconds (ms).
- Lower latency indicates faster communication.

Components of Latency

Latency consists of several types of delays:

- Transmission Delay : Time required to place all bits of data onto the transmission medium.
- Propagation Delay : Time taken for the signal to travel through the physical medium from sender to receiver.
- Queuing Delay : Time a packet waits in a queue before being transmitted due to network congestion.
- Processing Delay : Time required by network devices (routers/switches) to process packet information.

Effects of Latency

- Slow response in communication
- Delay in video calls and online gaming
- Increased waiting time for data transfer

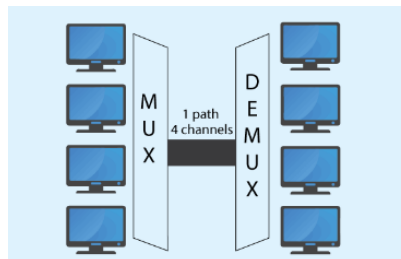
Example

During a video call, if there is a noticeable delay between speaking and hearing the response, it is caused by high latency.

❖ Multiplexing

Multiplexing is the technique of combining multiple signals or data streams into a single signal for transmission over a shared communication medium.

To achieve efficient utilization of bandwidth and communication resources, reduce cost, and allow multiple users to share the same channel.

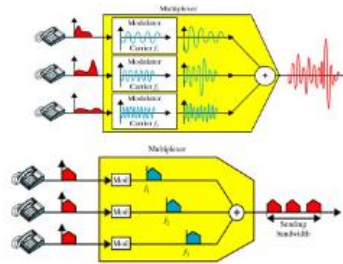


Types of Multiplexing:

1. Frequency Division Multiplexing (FDM)

- FDM involves dividing the available frequency spectrum into multiple non-overlapping frequency bands. Each signal is assigned a different frequency band for transmission.
- In the case of frequency division multiplexing, if the input signal is digital, it must be converted to analog before being used as the modulator's input.
- It is simple to send a huge number of signals at the same time.
- FDM multiplexing demodulation is simple.
- When employing FDM, there is a problem called crosstalk.
- When wideband fading occurs, it affects all channels in the FDM.

The FDM Process: Multiplexing



- Example: Analog radio broadcasting, where different radio stations transmit on distinct frequency bands within the allocated spectrum.

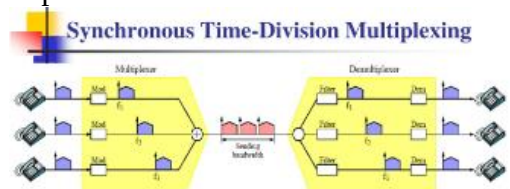
2. Time Division Multiplexing (TDM)

- TDM allocates fixed time slots to multiple signals within a single transmission cycle. Each signal takes turns to transmit during its designated time slot.
- The channel/link is separated by time rather than the frequency in this strategy.
- The total available time on the channel is shared among the many channel users.
- Each user on the channel is assigned a time slot/slice, which is a specific time interval.
- Example: T1/E1 digital telephone systems, where multiple phone conversations are combined into a single transmission line by assigning each conversation a time slot.

2.1. Types of TDM-multiplexing

a. Synchronous Time-Division Multiplexing (STDM)

- In this, multiplexer allocates same time slots to each device without considering fact that device contains data or not.
- Number of slots per frame are equal to number of input lines i.e., If it contains n input lines, then it must have n slots in one frame.
- There is no guarantee that full capacity link is used.
- Total speed of input lines cannot be greater than capacity of path.
- Number of time slots in a frame as always based on number of input lines.
- Time slots are fixed and pre-defined.



b. Asynchronous Time-Division

- In this, multiplexer does not allocates same time slots to each device without considering fact that device contains data or not.
- There is guarantee that full capacity link is used.
- Number of time slots in a frame as always based on statistical analysis of number of input lines that are likely to be transmitting at any given time.

2.3. Wavelength Division Multiplexing (WDM)

- WDM uses different wavelengths of light to carry multiple data streams simultaneously over optical fibers. Each wavelength represents a separate channel.

WDM is same like FDM

Example: Fiber optic networks, where multiple data signals are transmitted over the same fiber using different wavelengths of light.

❖ **Transmission Media**

Transmission media are the physical pathways through which data is transmitted from one device to another in a network. They carry signals in the form of electrical pulses, light, or electromagnetic waves.

Types of Transmission Media

Transmission media are broadly classified into:

1. Guided Media (Wired Media)
2. Unguided Media (Wireless Media)

1. Guided Media

- These media provide a physical path for signals to travel.
- Physical pathways that guide data signals from sender to receiver.

Types of Guided Media:

- a) Twisted Pair Cable
- b) Coaxial Cable
- c) Fiber Optic Cable

a) Twisted Pair Cable

Twisted pair cable is one of the most commonly used guided transmission media. It consists of two insulated copper wires twisted together in pairs. The twisting reduces electromagnetic interference (EMI) and crosstalk (interference from nearby wires). Twisted pair cables are widely used in telephone networks and Local Area Networks (LANs).

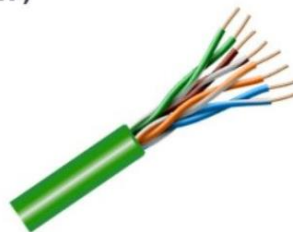


Types of Twisted Pair Cable

1. Unshielded Twisted Pair (UTP)

Unshielded Twisted Pair (UTP) cable is the most widely used network cable, especially in Local Area Networks (LANs). It consists of four pairs of insulated copper wires twisted together without any metallic shielding. Twisting reduces crosstalk and electromagnetic interference (EMI). UTP cables are classified into different categories (Cat) based on their bandwidth, data rate, and performance.

Unshielded Twisted Pair (UTP)



UTP cables are classified into categories (Cat) based on their speed, bandwidth, and performance:

Category	Data Rate	Bandwidth	Use
Cat 1	Voice only	Very low	Telephone
Cat 2	Up to 4 Mbps	Low	Old networks (obsolete)
Cat 3	Up to 10 Mbps	16 MHz	Early Ethernet (10Base-T), telephone
Cat 4	Up to 16 Mbps	20 MHz	Token Ring networks (rarely used)
Cat 5	Up to 100 Mbps	100 MHz	Fast Ethernet (100Base-T)

Category	Data Rate	Bandwidth	Use
Cat 5e	Up to 1 Gbps	100 MHz	Improved Cat 5, widely used in LANs
Cat 6	1 Gbps (100 m), 10 Gbps (55 m)	250 MHz	High-speed LANs
Cat 6a	10 Gbps (100 m)	500 MHz	Enterprise networks
Cat 7	10 Gbps+	600–1000 MHz	Data centers, high-speed networks

2. Shielded Twisted Pair (STP)

- Contains a metallic foil or braided shield around the twisted pairs.
- Provides better protection against EMI and external noise.
- Used in high-interference environments, such as factories, industrial areas, and heavy equipment zones.
- Slightly more expensive and less flexible than UTP.

Advantages of Twisted Pair Cable

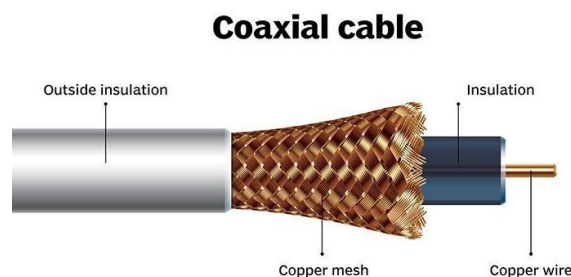
- Economical and easy to install
- Lightweight and flexible
- Reduces interference and crosstalk compared to straight wires

Disadvantages

- Limited transmission distance (about 100 meters for UTP)
- Lower bandwidth compared to coaxial and fiber optic cables
- STP is costlier and harder to install than UTP

b) Coaxial Cable

Coaxial cable is a guided transmission medium used in data communication. It is called coaxial because the central conductor and the outer metallic shield share the same axis. It provides better protection against electromagnetic interference (EMI) and supports higher bandwidth than twisted pair cable. Coaxial cables are commonly used in cable TV, internet connections, and CCTV systems.



RG Types of Coaxial Cables

RG (Radio Guide) cables are standard types of coaxial cables used for different communication purposes. They vary based on impedance, thickness, signal loss, and application.

1. RG-8

- Thick coaxial cable with 50-ohm impedance
- Used for high-power radio frequency (RF) transmission
- Provides low signal loss
- Commonly used in radio transmitters and antenna systems

2. RG-9

- Coaxial cable with 75-ohm impedance
- Designed for high-frequency and long-distance transmission
- Provides low attenuation
- Used in CATV and broadband applications

3. RG-11

- Thick coaxial cable with 75-ohm impedance
- Supports long-distance signal transmission

- Lower signal loss than RG-6 and RG-59
- Used in cable TV, satellite TV, and long cable runs

4. RG-58

- Thin coaxial cable with 50-ohm impedance
- Flexible and easy to install
- Used in Ethernet networks (10Base2) and RF applications
- Suitable for short-distance communication

5. RG-59

- Thin coaxial cable with 75-ohm impedance
- Used for low-frequency video signals
- Commonly used in CCTV and analog video systems
- Not suitable for long distances due to higher attenuation

Applications of Coaxial Cable

Coaxial cables are widely used in various communication systems due to their reliability and noise resistance:

- Cable television (TV) distribution
- Broadband internet connections-
- CCTV surveillance systems
- Radio transmitters and receivers
- Older Ethernet and LAN networks
- Satellite communication systems

Advantages of Coaxial Cable

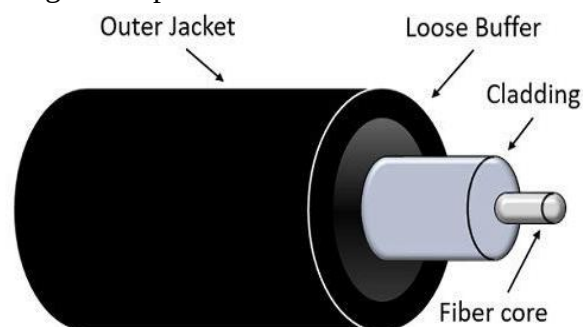
- Provides greater bandwidth than twisted pair cable
- Highly resistant to electromagnetic interference (EMI)
- Supports longer transmission distances
- Better signal quality and reduced noise
- Strong and durable construction

Disadvantages of Coaxial Cable

- Bulkier and less flexible than twisted pair cable
- More expensive than UTP cables
- Difficult to install and maintain
- Limited bandwidth compared to fiber optic cables

c) Fiber Optic Cable

- An optical fiber is a new transmission medium used for long-distance communication. It consists of a very thin glass or plastic cable that carries data in the form of light signals.



- An optical fiber cable is divided into three main parts:
 1. Core

- The core is the inner part of the optical fiber.
- It is made of glass or plastic.
- Light signals travel through the core.
- LED or laser light is used to transmit data.
- The core is extremely thin, almost as thin as a human hair.

2. Cladding

- The cladding surrounds the core.
- It is made of a glass-like material with a lower refractive index than the core.
- Its main function is to reflect light back into the core, preventing loss of signal.
- This reflection helps light travel long distances inside the fiber.

3. Protective Coating (Secondary Buffer)

- This is the outer covering of the optical fiber.
- It protects the fiber from damage, moisture, and bending.
- It increases the durability of the whole cable.

Refraction

- When a light ray passes from one transparent medium to another, its direction changes.
- This bending of light is called refraction of light.

Critical Angle

- When light travels from a denser medium to a rarer medium, there is a specific angle of incidence at which the angle of refraction becomes 90 degrees.
- This angle is called the critical angle.

Total Internal Reflection

- If a light ray strikes the boundary between two media at an angle greater than the critical angle,
- The light is completely reflected back into the denser medium instead of passing out.
- This phenomenon is called total internal reflection.

Working of Optical Fiber

- Optical fibers work on the principle of total internal reflection.
- The core has a higher refractive index than the cladding.
- Light entering the core strikes the core-cladding boundary at an angle greater than the critical angle.
- As a result, the light is continuously reflected inside the core and travels long distances without loss.

Advantages of Opt

1. High Speed:Optical fiber provides very high data transmission speed.
2. Large Bandwidth:It can carry a large amount of data at the same time.
3. Long Distance Communication:Signals can travel very long distances with very little loss.
4. Low Signal Loss:Less attenuation compared to copper cables.
5. Immunity to Electromagnetic Interference:Not affected by electrical noise, lightning, or radio interference.
6. Small Size and Lightweight:Optical fibers are thin like hair and easy to install.
7. High Security:Difficult to tap, so data transmission is more secure.
8. Corrosion Resistant:Does not corrode like metal cables.

Disadvantages of Optical Fiber

1. High Initial Cost:Installation and equipment are expensive.

2. **Fragile:**Optical fibers can break easily if bent too much.
3. **Difficult to Repair:**Requires skilled technicians and special tools.
4. **Complex Installation:**Splicing and joining fibers is difficult.
5. **Special Equipment Required:**Needs optical transmitters, receivers, and amplifiers.
6. **Not Suitable for Short Distance:**For short distances, copper cables may be cheaper and easier.

❖ Switching

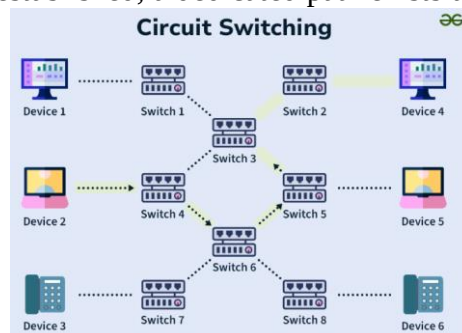
Switching is a technique used to transfer data from one device to another in a computer network.

Types of Switching

1. Circuit Switching
2. Packet Switching
 - a. Virtual Circuit Packet Switching
 - b. Datagram Packet Switching
3. Message Switching

Circuit Switching

- A circuit-switched network consists of a set of switches connected by physical links.
- Circuit switching works in a similar way to the telephone system.
- In this type of switching, two nodes communicate using a dedicated communication path.
- A pre-specified route is established before data transmission starts.
- No other data is allowed to use this path during communication.
- Before communication begins, resources are reserved for the entire duration of transmission.
- Once the connection is established, a dedicated path exists until the communication ends.



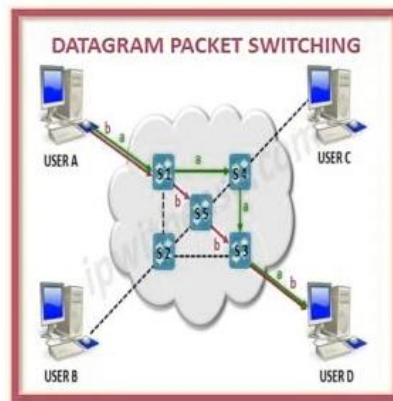
Example:

- Telephone network

2. Packet Switching Methods

a. Datagram Packet Switching

- Datagram packet switching is usually implemented at the Network Layer.
- In a datagram network, each packet is routed independently.
- Each packet carries a header containing the complete destination address.
- When a switch receives a packet, it:
 - Examines the destination address in the header.
 - Consults the routing table.
 - Forwards the packet through the appropriate outgoing port.
- Packets may take different paths and may reach the destination out of order.

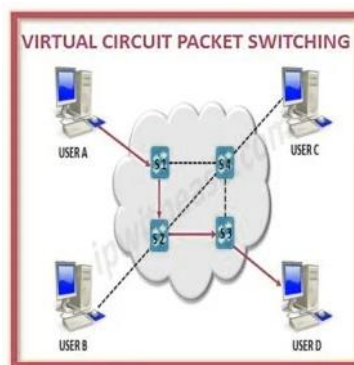


Example:

- Internet (IP)

b. Virtual Circuit Packet Switching

- Virtual circuit packet switching is normally implemented at the Data Link Layer.
- It establishes a fixed logical path between the source and destination.
- It is also known as a connection-oriented network.



- Communication occurs in three phases:
- Phases:
 - Setup Phase – Logical connection is established.
 - Data Transfer Phase – Packets are transmitted.
 - Connection Release Phase – Connection is terminated.
 - A logical connection is created when:
 - The sender sends a setup request.
 - The receiver sends an acknowledgement if it agrees.
- All packets between the same source and destination follow the same path.
- Data is delivered to the receiver in the same order as sent.

Example:

- Frame Relay
- ATM

3. Message Switching

- In message switching, there is no need to establish a dedicated path.
- Each message is routed independently through the network.
- Every message contains a header with complete destination information.
- Each intermediate device receives the entire message, stores it temporarily, and then forwards it.
- If the next node does not have enough resources, the message is stored and waits.
- Due to this, message switching is also called Store-and-Forward Switching.
- Message switching is very slow because of buffering delays.
- It is not suitable for real-time applications such as voice and video transmission.

❖ Introduction to Networking Devices: Repeater, Amplifier, HUB, Switches, Bridges, Routers

Internetworking devices are hardware devices used to connect two or more computers or networks for sharing data and resources. These devices operate at different layers of the OSI model and help in data transmission, filtering, and routing.

Hub

- A Hub is a multiport repeater used to connect multiple devices in a network.

Working:

- It receives data from one port
- Sends the same data to all other ports
- It does not filter data
- All connected devices receive the transmitted data

Characteristics:

- Operates at Physical Layer
- Not an intelligent device
- Does not select the best path
- Causes more network traffic
- It is a half-duplex device

Types of Hub:

- Active Hub
 - Regenerates and amplifies signals
 - Also called multiport repeater
- Passive Hub
 - Only connects devices
 - Does not amplify signals

Repeater

- A Repeater is a device used to extend the length of a network by regenerating weak signals.

Working:

- Regenerates weak signals
- Connects two segments of the same network
- Forwards every frame
- Has no filtering capability

Characteristics:

- Operates at Physical Layer
- It is a regenerator, not an amplifier
- Improves signal quality

Switch

- A Switch is an intelligent networking device that forwards data only to the destination device.

Working:

- Uses MAC addresses
- Sends data only to the required destination port
- Reduces unnecessary network traffic

Characteristics:

- Operates at Data Link Layer
- Supports full-duplex communication
- Faster than hub
- Commonly used in LANs

- Connects computers, printers, and servers

Bridge

- A Bridge is used to divide a network into segments and filter traffic.

Working:

- Checks MAC addresses of frames
- Uses a filtering table
- If destination is in the same segment, the packet is dropped
- Otherwise, the packet is forwarded to another segment

Characteristics:

- Operates at Physical and Data Link Layer
- Regenerates signals
- Reduces network collisions
- Slower than switch

Router

- A Router is an intelligent device that connects two or more different networks.

Working:

- Uses logical (IP) addresses
- Selects the best path using a routing table
- Routes packets from host to host

Characteristics:

- Operates at Network Layer
- Used in LAN, WAN, and Internet
- More intelligent than switch
- More expensive than hub and switch

Gateway

- A Gateway connects networks that use different protocols.

Working:

- Acts as a protocol converter
- Translates data between dissimilar networks
- Example: Email protocol conversion

Characteristics:

- Operates at Application Layer
- Most intelligent networking device
- Used to connect LAN to Internet
- Also called protocol converter

❖ **IMP QUESTIONS:**

1. **Multiplexing types (FDM, TDM, WDM)**
2. **Transmission Media comparison**
3. **Switching techniques**
4. **Transmission impairments**
5. **Analog vs Digital signals**
6. **Networking devices explanation**
7. **Network performance parameters**